

[illegible]

|             |                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Name</b> | Intro to EyeWitness                                                                                                   |
| <b>URL</b>  | <a href="https://attackdefense.com/challengedetails?cid=2115">https://attackdefense.com/challengedetails?cid=2115</a> |
| <b>Type</b> | Network Pentesting: Recon                                                                                             |

**Important Note:** This document illustrates all the important steps required to complete this lab. This is by no means a comprehensive step-by-step solution for this exercise. This is only provided as a reference to various commands needed to complete this exercise and for your further research on this topic. Also, note that the IP addresses and domain names might be different in your lab.

**Step 1:** Check the IP address of the machine.

**Command:** ip a

```
root@attackdefense:~#  
root@attackdefense:~# ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
66211: eth0@if66212: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default  
    link/ether 02:42:ac:11:00:03 brd ff:ff:ff:ff:ff:ff link-netnsid 0  
    inet 10.1.1.19/24 brd 10.1.1.255 scope global eth0  
        valid_lft forever preferred_lft forever  
66213: eth1@if66214: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default  
    link/ether 02:42:c0:6a:6e:02 brd ff:ff:ff:ff:ff:ff link-netnsid 0  
    inet 192.106.110.2/24 brd 192.106.110.255 scope global eth1  
        valid_lft forever preferred_lft forever  
root@attackdefense:~#
```

**Step 2:** Scanning the target machines using nmap.

**Command:** nmap 192.106.110.3-4

```
root@attackdefense:~#  
root@attackdefense:~# nmap 192.106.110.3-4  
Starting Nmap 7.70 ( https://nmap.org ) at 2020-07-31 16:48 IST  
Nmap scan report for target-1 (192.106.110.3)  
Host is up (0.000017s latency).  
Not shown: 999 closed ports  
PORT      STATE SERVICE  
8080/tcp  open  http-proxy  
MAC Address: 02:42:C0:6A:6E:03 (Unknown)  
  
Nmap scan report for target-2 (192.106.110.4)  
Host is up (0.000017s latency).  
Not shown: 998 closed ports  
PORT      STATE SERVICE  
80/tcp    open  http  
3306/tcp  open  mysql  
MAC Address: 02:42:C0:6A:6E:04 (Unknown)  
  
Nmap done: 2 IP addresses (2 hosts up) scanned in 0.26 seconds  
root@attackdefense:~#
```

On the first target machine, a service is running on port 8080 and on the second target machine, 2 services are running on port 80 (possibly an HTTP server) and port 3306 (possibly mysql).

**Step 3:** Using the EyeWitness tool to analyze the webapp running on the first target machine.

Using EyeWitness tool to scan a single host:

**Commands:**

```
cd EyeWitness/Python/  
./EyeWitness.py --single 192.106.110.3:8080
```

```
#####
#                               EyeWitness                               #
#####
#       FortyNorth Security - https://www.fortynorthsecurity.com       #
#####

Attempting to screenshot http://192.106.110.3:8080

[*] Done! Report written in the /root/EyeWitness/Python/07312020_164922 folder!
Would you like to open the report now? [Y/n]

```

The screenshot shows a web browser displaying the EyeWitness Report. The report is titled "Report Generated on 07/31/2020 at 16:49:22". It is divided into two main sections: "Web Request Info" and "Web Screenshot".

**Web Request Info:**

- URL: <http://192.106.110.3:8080>
- Default credentials: Apache Tomcat tomcat/tomcat admin/admin etc.
- Page Title: Apache Tomcat/9.0.37
- Content-Type: text/html; charset=UTF-8
- Transfer-Encoding: chunked
- Date: Fri, 31 Jul 2020 11:19:30 GMT
- Connection: close
- Response Code: 200
- Source Code

**Web Screenshot:**

The screenshot shows the Apache Tomcat 9.0.37 homepage. It includes a navigation bar with links like Home, Documentation, Configuration, Examples, Wiki, and Mailing Lists. The main content area features a green banner with the Tomcat logo and a message: "If you're seeing this, you've successfully installed Tomcat. Congratulations!". Below this, there are sections for "Recommended Reading" (Security Considerations, Manager Application, Clustering/Session Replication), "Developer Quick Start", "Managing Tomcat", "Documentation" (Tomcat 9.0 Documentation, Tomcat 9.0 Configuration, Tomcat Wiki), and "Getting Help" (FAQ and Mailing Lists, Tomcat announcements, Tomcat user support, Tomcat day, Development mailing list).

The tool has taken the screenshot of the webapp (default webapp of Apache Tomcat) running on the target machine and identified the default credentials for the webapp as well.

**Step 4:** Scanning both the target webapps using the EyeWitness tool.

Save the following endpoints to a file called urls.txt:

```
192.106.110.3:8080
192.106.110.4
```

**Command:** cat urls.txt

```
root@attackdefense:~/EyeWitness/Python#  
root@attackdefense:~/EyeWitness/Python# cat urls.txt  
192.106.110.3:8080  
192.106.110.4  
root@attackdefense:~/EyeWitness/Python#
```

Use the following command to take the screenshots of the specified URLs and identify if it could find any default credentials for those webapps.

**Command:** ./EyeWitness.py -f urls.txt --web

```
#####  
#                               EyeWitness                               #  
#####  
#       FortyNorth Security - https://www.fortynorthsecurity.com       #  
#####  
  
Starting Web Requests (2 Hosts)  
Attempting to screenshot http://192.106.110.3:8080  
Attempting to screenshot http://192.106.110.4  
[*] Hit timeout limit when connecting to http://192.106.110.4, retrying  
[*] Hit timeout limit when connecting to http://192.106.110.4  
Finished in 33.77864146232605 seconds  
  
[*] Done! Report written in the /root/EyeWitness/Python/07312020_165131 folder!  
Would you like to open the report now? [Y/n]  
█
```

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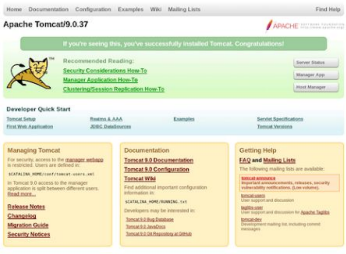
|                    |   |
|--------------------|---|
| High Value Targets | 1 |
| Errors             | 1 |
| Total              | 2 |

Report Generated on 07/31/2020 at 16:51:31

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## High Value Targets

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                               | Web Screenshot                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <p><a href="http://192.106.110.3:8080">http://192.106.110.3:8080</a></p> <p>Default credentials: Apache Tomcat<br/>tomcat/tomcat admin/admin etc.</p> <p>Page Title: Apache Tomcat/9.0.37</p> <p>Content-Type: text/html; charset=UTF-8</p> <p>Transfer-Encoding: chunked</p> <p>Date: Fri, 31 Jul 2020 11:21:34 GMT</p> <p>Connection: close</p> <p>Response Code: 200</p> <p><a href="#">Source Code</a></p> |  |

Report Generated on 07/31/2020 at 16:51:31

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## Errors

| Web Request Info                                                                            | Web Screenshot                                          |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <p><a href="http://192.106.110.4">http://192.106.110.4</a></p> <p>Resolved to: target-2</p> | <p>Hit timeout limit while attempting to screenshot</p> |

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The second URL results in a timeout. Hence, its screenshot couldn't be taken.

**Eyewitness uses a default timeout of 7 seconds. It can be increased by using the `--timeout` flag.**

Scanning the target URLs using an increased timeout value. Also restricting the maximum number of retries to 1:

**Command:** `./Eyewitness.py -f urls.txt --web --timeout 45 --max-retries 1`

```
#####
#                               EyeWitness                               #
#####
#          FortyNorth Security - https://www.fortynorthsecurity.com      #
#####

Starting Web Requests (2 Hosts)
Attempting to screenshot http://192.106.110.3:8080
Attempting to screenshot http://192.106.110.4
Finished in 47.72801947593689 seconds

[*] Done! Report written in the /root/EyeWitness/Python/07312020_165612 folder!
Would you like to open the report now? [Y/n]

```

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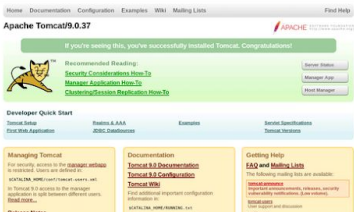
|                    |   |
|--------------------|---|
| High Value Targets | 1 |
| Uncategorized      | 1 |
| Errors             | 0 |
| Total              | 2 |

Report Generated on 07/31/2020 at 16:56:12

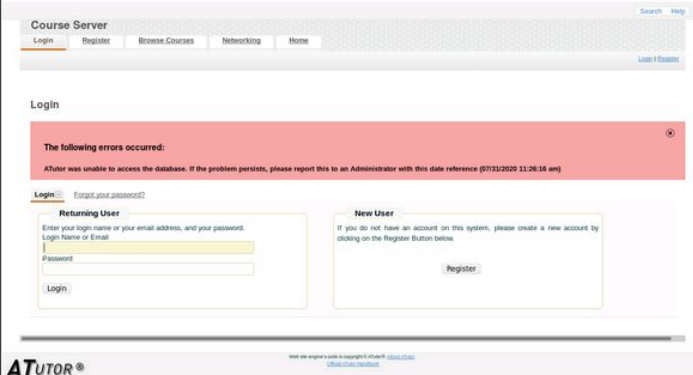
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## High Value Targets

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                              | Web Screenshot                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p><a href="http://192.106.110.3:8080">http://192.106.110.3:8080</a></p> <p><b>Default credentials:</b> Apache Tomcat<br/>tomcat/tomcat admin/admin etc.</p> <p><b>Page Title:</b> Apache Tomcat/9.0.37<br/><b>Content-Type:</b> text/html; charset=UTF-8<br/><b>Transfer-Encoding:</b> chunked<br/><b>Date:</b> Fri, 31 Jul 2020 11:26:15 GMT<br/><b>Connection:</b> close<br/><b>Response Code:</b> 200</p> |  |

## Uncategorized

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Web Screenshot                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <a href="http://192.106.110.4">http://192.106.110.4</a><br><b>Resolved to:</b> target-2<br><br><b>Default credentials:</b> (ATutor Web CMS)<br>admin/admin or instructor:instructor<br><br><b>Page Title:</b> Course Server: Login<br><b>Date:</b> Fri, 31 Jul 2020 11:26:57 GMT<br><b>Server:</b> Apache/2.4.7 (Ubuntu)<br><b>X-Powered-By:</b> PHP/5.5.9-1ubuntu4.25<br><b>Set-Cookie:</b><br>ATutorID=kbsfs31fph9el26b98pl7e6u45;<br>path=/<br><b>P3P:</b> CP="IDC DSP COR CURa ADMa OUR IND PHY ONL COM STA"<br><b>Vary:</b> Accept-Encoding<br><b>Connection:</b> close<br><b>Transfer-Encoding:</b> chunked<br><b>Content-Type:</b> text/html; charset=utf-8<br><b>Response Code:</b> 200<br><br><a href="#">Source Code</a> |  |

This time the tool had successfully taken the screenshot of the second webapp and also identified the default credentials for it from its signatures.

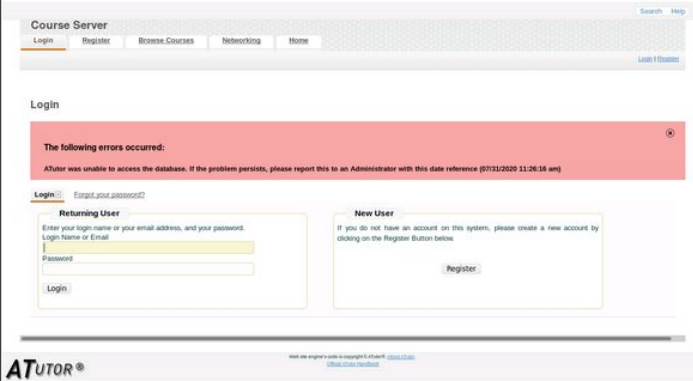
By default, EyeWitness creates a directory in the same folder where EyeWitness.py file is present and saves all the screenshots and reports in it.

**To save the screenshots in some other directory, use the -d flag:**

**Command:** ./EyeWitness.py -f urls.txt --web --timeout 45 -d /tmp/ew-out



## Uncategorized

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Web Screenshot                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <p><a href="http://192.106.110.4">http://192.106.110.4</a><br/> <b>Resolved to:</b> target-2</p> <p><b>Default credentials:</b> (ATutor Web CMS)<br/> admin/admin or instructor:instructor</p> <p><b>Page Title:</b> Course Server: Login<br/> <b>Date:</b> Fri, 31 Jul 2020 11:26:57 GMT<br/> <b>Server:</b> Apache/2.4.7 (Ubuntu)<br/> <b>X-Powered-By:</b> PHP/5.5.9-1ubuntu4.25<br/> <b>Set-Cookie:</b><br/> ATutorID=kbsfs31fph9el26b98pl7e6u45;<br/> path=/<br/> <b>P3P:</b> CP="IDC DSP COR CURa ADMa<br/> OUR IND PHY ONL COM STA"<br/> <b>Vary:</b> Accept-Encoding<br/> <b>Connection:</b> close<br/> <b>Transfer-Encoding:</b> chunked<br/> <b>Content-Type:</b> text/html; charset=utf-8<br/> <b>Response Code:</b> 200</p> <p><a href="#">Source Code</a></p> |  |

**EyeWitness can also be used to resolve the IP addresses to hostnames and vice-versa.**

**Command:** ./EyeWitness.py --resolve --single 192.106.110.4 --timeout 3

```
#####
#                               EyeWitness                               #
#####
#       FortyNorth Security - https://www.fortynorthsecurity.com       #
#####

Attempting to screenshot http://192.106.110.4
[*] Hit timeout limit when connecting to http://192.106.110.4, retrying
[*] Hit timeout limit when connecting to http://192.106.110.4

[*] Done! Report written in the /root/EyeWitness/Python/07312020_170235 folder!
Would you like to open the report now? [Y/n]
█
```

EyeWitness Report

file:///root/EyeWitness/Python/07312020\_170235/report.html

Report Generated on 07/31/2020 at 17:02:35

| Web Request Info                                                                 | Web Screenshot                                   |
|----------------------------------------------------------------------------------|--------------------------------------------------|
| <a href="http://192.106.110.4">http://192.106.110.4</a><br>Resolved to: target-2 | Hit timeout limit while attempting to screenshot |

The hostname of the second target machine (IP address 192.106.110.4) is “target-2”.

**EyeWitness can read the hosts to be scanned from the XML output of nmap as well:**

**Command:** nmap --top-ports 20 192.106.110.0/28 -oX out.xml

```
root@attackdefense:~/EyeWitness/Python# nmap --top-ports 20 192.106.110.0/28 -oX out.xml
Starting Nmap 7.70 ( https://nmap.org ) at 2020-07-31 17:03 IST
Nmap scan report for 192.106.110.1
Host is up (0.000015s latency).
```

| PORT    | STATE    | SERVICE     |
|---------|----------|-------------|
| 21/tcp  | closed   | ftp         |
| 22/tcp  | open     | ssh         |
| 23/tcp  | closed   | telnet      |
| 25/tcp  | closed   | smtp        |
| 53/tcp  | closed   | domain      |
| 80/tcp  | filtered | http        |
| 110/tcp | closed   | pop3        |
| 111/tcp | closed   | rpcbind     |
| 135/tcp | closed   | msrpc       |
| 139/tcp | closed   | netbios-ssn |
| 143/tcp | closed   | imap        |

...

|          |        |               |
|----------|--------|---------------|
| 21/tcp   | closed | ftp           |
| 22/tcp   | closed | ssh           |
| 23/tcp   | closed | telnet        |
| 25/tcp   | closed | smtp          |
| 53/tcp   | closed | domain        |
| 80/tcp   | closed | http          |
| 110/tcp  | closed | pop3          |
| 111/tcp  | closed | rpcbind       |
| 135/tcp  | closed | msrpc         |
| 139/tcp  | closed | netbios-ssn   |
| 143/tcp  | closed | imap          |
| 443/tcp  | closed | https         |
| 445/tcp  | closed | microsoft-ds  |
| 993/tcp  | closed | imaps         |
| 995/tcp  | closed | pop3s         |
| 1723/tcp | closed | pptp          |
| 3306/tcp | closed | mysql         |
| 3389/tcp | open   | ms-wbt-server |
| 5900/tcp | closed | vnc           |
| 8080/tcp | closed | http-proxy    |

```
Nmap done: 16 IP addresses (4 hosts up) scanned in 14.73 seconds
root@attackdefense:~/EyeWitness/Python#
```

Checking the generating XML output:

**Command:** vim out.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE nmaprun>
<?xml-stylesheet href="file:///usr/bin/./share/nmap/nmap.xsl" type="text/xsl"?>
<!-- Nmap 7.70 scan initiated Fri Jul 31 17:03:57 2020 as: nmap -6#45;top-ports 20 -oX out.xml 192.106.110.0/28 -->
<nmaprun scanner="nmap" args="nmap -6#45;top-ports 20 -oX out.xml 192.106.110.0/28" starttime="1596195237" startstr="Fri Jul 31 17:03:57 2020" version="7.70" xmloutputversion="1.04">
<scaninfo type="syn" protocol="tcp" numservices="20" services="21-23,25,53,80,110-111,135,139,143,443,445,993,995,1723,3306,3389,5900,8080"/>
<verbose level="0"/>
<debugging level="0"/>
<host starttime="1596195237" endtime="1596195251"><status state="up" reason="arp-response" reason_ttl="0"/>
<address addr="192.106.110.1" addrtype="ipv4"/>
<address addr="02:42:AB:32:AC:FB" addrtype="mac"/>
<hostnames>
</hostnames>
<ports>
<port protocol="tcp" portid="21"><state state="closed" reason="reset" reason_ttl="64"/><service name="ftp" method="table" conf="3"/></port>
<port protocol="tcp" portid="22"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="ssh" method="table" conf="3"/></port>
<port protocol="tcp" portid="23"><state state="closed" reason="reset" reason_ttl="64"/><service name="telnet" method="table" conf="3"/></port>
<port protocol="tcp" portid="25"><state state="closed" reason="reset" reason_ttl="64"/><service name="smtp" method="table" conf="3"/></port>
<port protocol="tcp" portid="53"><state state="closed" reason="reset" reason_ttl="64"/><service name="domain" method="table" conf="3"/></port>
<port protocol="tcp" portid="80"><state state="filtered" reason="no-response" reason_ttl="0"/><service name="http" method="table" conf="3"/></port>
<port protocol="tcp" portid="110"><state state="closed" reason="reset" reason_ttl="64"/><service name="pop3" method="table" conf="3"/></port>
<port protocol="tcp" portid="111"><state state="closed" reason="reset" reason_ttl="64"/><service name="rpcbind" method="table" conf="3"/></port>
<port protocol="tcp" portid="135"><state state="closed" reason="reset" reason_ttl="64"/><service name="msrpc" method="table" conf="3"/></port>
<port protocol="tcp" portid="139"><state state="closed" reason="reset" reason_ttl="64"/><service name="netbios-ssn" method="table" conf="3"/></port>
<port protocol="tcp" portid="143"><state state="closed" reason="reset" reason_ttl="64"/><service name="imap" method="table" conf="3"/></port>
<port protocol="tcp" portid="443"><state state="closed" reason="reset" reason_ttl="64"/><service name="https" method="table" conf="3"/></port>
<port protocol="tcp" portid="445"><state state="closed" reason="reset" reason_ttl="64"/><service name="microsoft-ds" method="table" conf="3"/></port>
<port protocol="tcp" portid="993"><state state="closed" reason="reset" reason_ttl="64"/><service name="imaps" method="table" conf="3"/></port>
<port protocol="tcp" portid="995"><state state="closed" reason="reset" reason_ttl="64"/><service name="pop3s" method="table" conf="3"/></port>
<port protocol="tcp" portid="1723"><state state="closed" reason="reset" reason_ttl="64"/><service name="pptp" method="table" conf="3"/></port>
<port protocol="tcp" portid="3306"><state state="closed" reason="reset" reason_ttl="64"/><service name="mysql" method="table" conf="3"/></port>
<port protocol="tcp" portid="3389"><state state="closed" reason="reset" reason_ttl="64"/><service name="ms-wbt-server" method="table" conf="3"/></port>
<port protocol="tcp" portid="5900"><state state="closed" reason="reset" reason_ttl="64"/><service name="vnc" method="table" conf="3"/></port>
<port protocol="tcp" portid="8080"><state state="closed" reason="reset" reason_ttl="64"/><service name="http-proxy" method="table" conf="3"/></port>
</ports>
```

Passing the XML output to EyeWitness tool:

**Command:** ./EyeWitness.py -x out.xml --web

```
#####
#                               EyeWitness                               #
#####
#   FortyNorth Security - https://www.fortynorthsecurity.com           #
#####

Starting Web Requests (2 Hosts)
Attempting to screenshot http://192.106.110.3:8080
Attempting to screenshot http://192.106.110.4
[*] Hit timeout limit when connecting to http://192.106.110.4, retrying
[*] Hit timeout limit when connecting to http://192.106.110.4
Finished in 33.77844214439392 seconds

[*] Done! Report written in the /root/EyeWitness/Python/07312020_170711 folder!
Would you like to open the report now? [Y/n]

```

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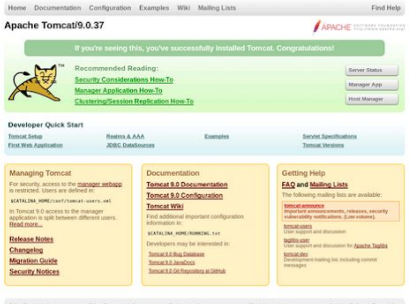
|                    |   |
|--------------------|---|
| High Value Targets | 1 |
| Errors             | 1 |
| Total              | 2 |

Report Generated on 07/31/2020 at 17:07:11

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## High Value Targets

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                                       | Web Screenshot                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <a href="http://192.106.110.3:8080">http://192.106.110.3:8080</a><br><b>Default credentials:</b> Apache Tomcat<br>tomcat/tomcat admin/admin etc.<br><b>Page Title:</b> Apache Tomcat/9.0.37<br><b>Content-Type:</b> text/html;charset=UTF-8<br><b>Transfer-Encoding:</b> chunked<br><b>Date:</b> Fri, 31 Jul 2020 11:37:14 GMT<br><b>Connection:</b> close<br><b>Response Code:</b> 200<br><a href="#">Source Code</a> |  |

file:///root/EyeWitness/Python/07312020\_170711/report\_page2.html

Report Generated on 07/31/2020 at 17:07:11

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## Errors

| Web Request Info                                                                        | Web Screenshot                                   |
|-----------------------------------------------------------------------------------------|--------------------------------------------------|
| <a href="http://192.106.110.4">http://192.106.110.4</a><br><b>Resolved to:</b> target-2 | Hit timeout limit while attempting to screenshot |

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**EyeWitness also provides the ability to specify a different user-agent string to be used while performing the target analysis:**

**Command:** `./EyeWitness.py --single 192.106.110.3:8080 --user-agent "Mozilla/4.0"`

```
#####
#                               EyeWitness                               #
#####
#       FortyNorth Security - https://www.fortynorthsecurity.com       #
#####

Attempting to screenshot http://192.106.110.3:8080

[*] Done! Report written in the /root/EyeWitness/Python/07312020_171003 folder!
Would you like to open the report now? [Y/n]
█
```

file:///root/EyeWitness/Python/07312020\_171003/report.html

Report Generated on 07/31/2020 at 17:10:03

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                                                                           | Web Screenshot |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <p><a href="http://192.106.110.3:8080">http://192.106.110.3:8080</a></p> <p><b>Default credentials:</b> Apache Tomcat tomcat/tomcat admin/admin etc.</p> <p><b>Page Title:</b> Apache Tomcat/9.0.37</p> <p><b>Content-Type:</b> text/html;charset=UTF-8</p> <p><b>Transfer-Encoding:</b> chunked</p> <p><b>Date:</b> Fri, 31 Jul 2020 11:40:06 GMT</p> <p><b>Connection:</b> close</p> <p><b>Response Code:</b> 200</p> <p><a href="#">Source Code</a></p> |                |

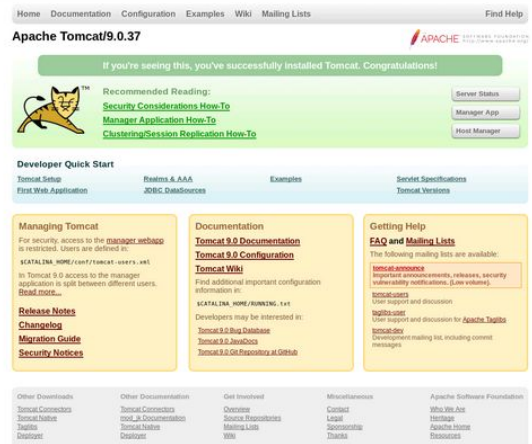
The prompt to open the report could be avoided using the **--no-prompt** flag:

**Command:** `./EyeWitness.py --single 192.106.110.3:8080 -d /tmp/ew-out --no-prompt`



file:///tmp/ew-out/report.html

Report Generated on 07/31/2020 at 17:11:13

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                                                                | Web Screenshot                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <p><a href="http://192.106.110.3:8080">http://192.106.110.3:8080</a></p> <p><b>Default credentials:</b> Apache Tomcat<br/>tomcat/tomcat admin/admin etc.</p> <p><b>Page Title:</b> Apache Tomcat/9.0.37<br/><b>Content-Type:</b> text/html;charset=UTF-8<br/><b>Transfer-Encoding:</b> chunked<br/><b>Date:</b> Fri, 31 Jul 2020 11:41:16 GMT<br/><b>Connection:</b> close<br/><b>Response Code:</b> 200</p> <p><a href="#">Source Code</a></p> |  |

A random delay between the requests sent to the target machine could be done using the --jitter flag:

Command: `./EyeWitness.py -f urls.txt --web --timeout 45 --jitter 10`

```
#####
#                               Eyewitness                               #
#####
#                               FortyNorth Security - https://www.fortynorthsecurity.com                               #
#####

Starting Web Requests (2 Hosts)
Attempting to screenshot http://192.106.110.3:8080
[*] Sleeping for 7.800000000000001 seconds..
Attempting to screenshot http://192.106.110.4
[*] Sleeping for 8.6 seconds..
Finished in 58.90510058403015 seconds

[*] Done! Report written in the /root/EyeWitness/Python/07312020_171401 folder!
Would you like to open the report now? [Y/n]
█
```

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- [Uncategorized \(Page 2\)](#)

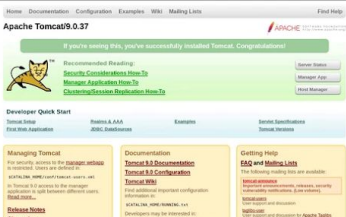
|                    |   |
|--------------------|---|
| High Value Targets | 1 |
| Uncategorized      | 1 |
| Errors             | 0 |
| Total              | 2 |

Report Generated on 07/31/2020 at 17:14:01

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## High Value Targets

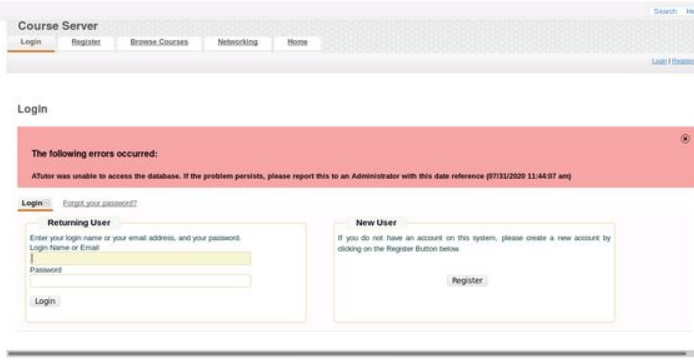
| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                              | Web Screenshot                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <p><a href="http://192.106.110.3:8080">http://192.106.110.3:8080</a></p> <p><b>Default credentials:</b> Apache Tomcat<br/>tomcat/tomcat admin/admin etc.</p> <p><b>Page Title:</b> Apache Tomcat/9.0.37<br/><b>Content-Type:</b> text/html; charset=UTF-8<br/><b>Transfer-Encoding:</b> chunked<br/><b>Date:</b> Fri, 31 Jul 2020 11:44:04 GMT<br/><b>Connection:</b> close<br/><b>Response Code:</b> 200</p> |  |

Report Generated on 07/31/2020 at 17:14:01

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## Uncategorized

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Web Screenshot                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p><a href="http://192.106.110.4">http://192.106.110.4</a><br/><b>Resolved to:</b> target-2</p> <p><b>Default credentials:</b> (ATutor Web CMS)<br/>admin/admin or instructor:instructor</p> <p><b>Page Title:</b> Course Server: Login<br/><b>Date:</b> Fri, 31 Jul 2020 11:44:48 GMT<br/><b>Server:</b> Apache/2.4.7 (Ubuntu)<br/><b>X-Powered-By:</b> PHP/5.5.9-1ubuntu4.25<br/><b>Set-Cookie:</b><br/>ATutorID=pst0qchfd5obf0bj8mjdlnib0;<br/>path=/<br/><b>P3P:</b> CP="IDC DSP COR CURa ADMa<br/>OUR IND PHY ONL COM STA"<br/><b>Vary:</b> Accept-Encoding<br/><b>Connection:</b> close<br/><b>Transfer-Encoding:</b> chunked<br/><b>Content-Type:</b> text/html; charset=utf-8<br/><b>Response Code:</b> 200</p> <p><a href="#">Source Code</a></p> |  |

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## EyeWitness Signatures:

EyeWitness tool comes with a set of signatures that it uses to identify the credentials of the target web application. These signatures are stored in signatures.txt file:

**Command:** ls

```
root@attackdefense:~/EyeWitness/Python#  
root@attackdefense:~/EyeWitness/Python# ls  
07312020_164922 07312020_170711 EyeWitness.py    bin          out.xml        urls.txt  
07312020_165131 07312020_171003 MiktoList.py     categories.txt parsed_xml.txt  
07312020_165612 07312020_171401 Recategorize.py  geckodriver.log setup  
07312020_170235 Dockerfile      Search.py        modules       signatures.txt  
root@attackdefense:~/EyeWitness/Python#
```

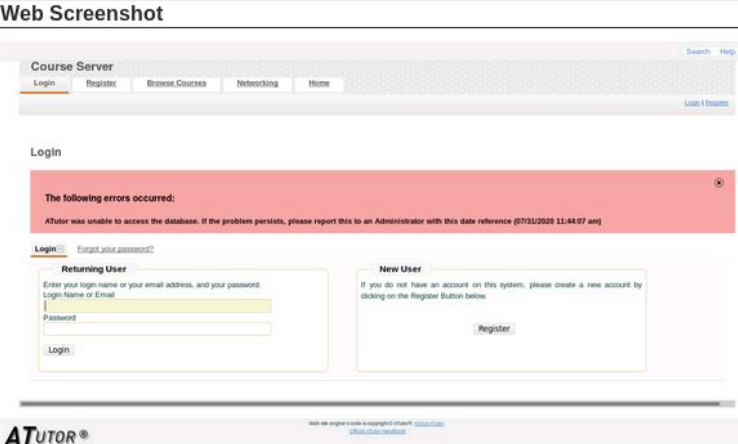
**Command:** vim signatures.txt

```
Integrated Dell Remote Access Controller|(Dell iDRAC) root/calvin  
ACE 4710 Device Manager;Cisco|(ACE 4710 Device Manager) admin/admin  
ATutor;form_login_action;/search.php|(ATutor Web CMS) admin/admin or instructor:instructor  
ATutor;form_login_action;/gad_search.php|(ATutor Web CMS) admin/admin or instructor:instructor  
Apache Lenya;lenya.action=introspect|(Apache Lenya) admin/levi  
Polycom RSS 2000|(Polycom RSS 2000) Administrator/polycom  
Polycom;HDX 8000 HD|(Polycom HDX 8000) admin/<SerialNumberofDevice>
```

Notice the signatures for the ATutor web app.

If all of the signatures listed for an entry match, then it is marked as detected and any default credentials displayed, otherwise only its screenshot is shown.

## Uncategorized

| Web Request Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Web Screenshot                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <p><a href="http://192.106.110.4">http://192.106.110.4</a><br/> <b>Resolved to:</b> target-2</p> <p><b>Default credentials:</b> (ATutor Web CMS)<br/> admin/admin or instructor:instructor</p> <p><b>Page Title:</b> Course Server: Login<br/> <b>Date:</b> Fri, 31 Jul 2020 11:44:48 GMT<br/> <b>Server:</b> Apache/2.4.7 (Ubuntu)<br/> <b>X-Powered-By:</b> PHP/5.5.9-1ubuntu4.25<br/> <b>Set-Cookie:</b><br/> ATutorID=pst0qchfd5obf0bj8mjdlcnib0;<br/> path=/<br/> <b>P3P:</b> CP="IDC DSP COR CURa ADMa<br/> OUR IND PHY ONL COM STA"<br/> <b>Vary:</b> Accept-Encoding<br/> <b>Connection:</b> close<br/> <b>Transfer-Encoding:</b> chunked<br/> <b>Content-Type:</b> text/html; charset=utf-8<br/> <b>Response Code:</b> 200</p> <p><a href="#">Source Code</a></p> |  |

Checking the source code of the second web application (ATutor):

```

<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en"><head>
  <title>Course Server: Login</title>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
  <meta name="Generator" content="ATutor - Copyright 2013 by http://atutor.ca">
  <base href="http://192.106.110.4/">
  <link rel="shortcut icon" href="http://192.106.110.4/favicon.ico" type="image/x-icon">
  <link rel="stylesheet" href="http://192.106.110.4/themes/default/print.css" type="text/css" media="print">
  <link rel="stylesheet" href="http://192.106.110.4/jscripits/infusion/framework/fss/css/fss-layout.css" type="text/css">
  <link rel="stylesheet" href="http://192.106.110.4/themes/default/styles.css" type="text/css">
  <link rel="stylesheet" href="http://192.106.110.4/themes/default/forms.css" type="text/css">
  <!--[if IE]>
    <link rel="stylesheet" href="http://192.106.110.4/themes/default/ie_styles.css" type="text/css" />
  <![endif]>
  <!--[if IE 8]>
    <link rel="stylesheet" href="http://192.106.110.4/themes/default/ie8_styles.css" type="text/css" />
  <![endif]>

  <!-- Fluid and jQuery Dependencies -->
  <script async=" " src="//www.google-analytics.com/analytics.js"></script><script type="text/javascript" src="http://192.106.110.
  progEnhance-enhanced, .fl-ProgEnhance-enhanced { display: block; }</style>
  <script type="text/javascript" src="http://192.106.110.4/jscripits/infusion/framework/enhancement/js/ProgressiveEnhancement.js">
  enhanced { display: block; }</style>
  <script src="/jscripits/mobile.js" type="text/javascript"></script>
  <script type="text/javascript">
    //<!--
    jQuery.noConflict();
    //-->
  </script>

```

The signature file looks for the following 3 strings in to match for ATutor:

- ATutor
- form\_login\_action
- /search.php

Press CTRL + F and look for all of these strings (case-insensitive match is done in the tool):

```

<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en"><head>
  <title>Course Server: Login</title>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
  <meta name="Generator" content="ATutor - Copyright 2013 by http://atutor.ca">
  <base href="http://192.106.110.4/">
  <link rel="shortcut icon" href="http://192.106.110.4/favicon.ico" type="image/x-icon">
  <link rel="stylesheet" href="http://192.106.110.4/themes/default/print.css" type="text/css" media="print">
  <link rel="stylesheet" href="http://192.106.110.4/jscripits/infusion/framework/fss/css/fss-layout.css" type="text/css">
  <link rel="stylesheet" href="http://192.106.110.4/themes/default/styles.css" type="text/css">
  <link rel="stylesheet" href="http://192.106.110.4/themes/default/forms.css" type="text/css">
  <!--[if IE]>

```

```

<div class="input-form">
  <div class="column" style="margin-top:0;">
    <form action="/login.php" method="post" name="form">
      <input type="hidden" name="form_login_action" value="true">
      <input type="hidden" name="form_course_id" value="0">
      <input type="hidden" name="form_password_hidden" value="">
      <input type="hidden" name="p" value="">

      <fieldset class="group_form"><legend class="group_form">Returning User</legend>
      <p>Enter your login name or your email address, and your password.</p>

      <label for="login">Login Name or Email</label><br>
      <input type="text" name="form_login" size="50" style="max-width: 80%; width: 80%;" id="login"><br>

      <label for="pass">Password</label><br>
      <input type="password" class="formfield" name="form_password" style="max-width: 80%; width: 80%;" id="pass">
      <br><br>
      <input type="submit" name="submit" value="Login" class="button" onclick="return encrypt_password();">
    </fieldset>
  </form>
</div>

<div class="site-name">&nbsp;</div>
<div id="top-links" role="navigation" aria-label="User">
  <!-- top help/search/login links -->

  <div id="top-links-text">

    <a href="/search.php">Search</a>
    <a href="/help/index.php">Help</a>

  </div>
</div>

```

All 3 strings were found in the page source and hence, the tool was able to identify it and display the default credentials for it.

## References:

1. EyeWitness (<https://github.com/FortyNorthSecurity/EyeWitness>)
2. EyeWitness Usage Guide (<https://www.christophertruncer.com/eyewitness-usage-guide/>)